

OGC JSON-FG

Geonovum Standards workshop

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Datum Wednesday 25 June 2026



Program



10.00 - 10.15	Welcome and introductions
10.15 - 10.30	Introduction to OGC in general
10.30 - 11.00	OGC JSON-FG in detail
11:00 - 11.15	Break
11.15 - 11.30	Validating a JSON-FG file
11.30 - 12.00	Hands-on session: Get started with JSON-FG yourself
12.00 - 12.45	Lunch break
12.45 - 13.30	Brainstorm pilot project with data exchange
13.30 - 14.45	Hands-on session: Get on with JSON-FG yourself
14.45 - 15.00	Wrap-up and closing
15.00 - 16.00	Optional extension: You are welcome to stay longer for further support from our API experts.

JSON FG is an OGC-standard

About The Open Geospatial Consortium (OGC):

- International standards organization for geospatial data and services
- Develops open, consensus-based standards to make geospatial information findable, accessible, interoperable and reusable
- Members include government agencies, companies, research institutes and universities worldwide
- Maintains widely used standards such as WMS, WFS, GeoPackage and the modern OGC API family

Latest News



Announcement • May 21, 2026

OGC Releases the Features and Geometries JSON (JSON-FG) Standard

[Read More](#) >



Announcement • May 12, 2026

OGC Publishes openEO as a New Community Standard

[Read More](#) >

Why is Geonovum supporting JSON-FG

- Request from Dutch Kadaster / Sweco
- Geonovum was involved in developing JSON-FG
- It contributes to a higher goal: encouraging the use of OGC API's, because API's benefit from this new standard.
- It will be used for new BGT data exchange.

What is JSON-FG

- Features and Geometries JSON
 - It is an encoding for special geometries
 - Extension to GeoJSON
 - Backwards compatible
-
- Established on April 30, 2026:
 - <https://www.ogc.org/standards/json-fg/>

<https://docs.ogc.org/is/21-045r1/21-045r1.html>



Open Geospatial Consortium

Submission Date: 2025-06-02

Approval Date: 2025-08-25

Publication Date: 2026-04-30

External identifier of this OGC® document: <http://www.opengis.net/doc/IS/json-fg-1/1.0>

Internal reference number of this OGC® document: 21-045r1

Version: 1.0.0

Category: OGC® Implementation Standard

Editor: Clemens Portele, Panagiotis (Peter) A. Vretanos

OGC Features and Geometries JSON - Part 1: Core

Why JSON-FG

- GeoJSON is widely used (OGC API's)
- But, it has its limitations:
 - only WGS84
 - no 3D
 - no standard for datetime attributes
 - no curves, arcs nor measures
 - no reference to schemas and feature type

JSON-FG has it!

Mandatory Extensions

- Ability to use Coordinate Reference Systems (**CRS**) other than WGS 84;
- Ability to encode **temporal** characteristics of a feature;
- A statement identifying the **JSON-FG conformance** classes supported by a JSON-FG feature collection, feature, or geometry.

JSON-FG has it!

Optional Extensions with support for:

- **3D:** solids and prisms as geometry types;
- **Complex geometry:**
 - circular arcs
 - compound curves
 - curve polygons;
- Support for **measure** values;
- The ability to declare the **type** and **schema** of a feature.

CRS

WGS84 (longitude, latitude)



RD (x, y)



Use **coordRefSys**-element:

- RD "coordRefSys": "<http://www.opengis.net/def/crs/EPSG/0/28992>",
- WGS84 "coordRefSys": "<http://www.opengis.net/def/crs/OGC/0/CRS84>",

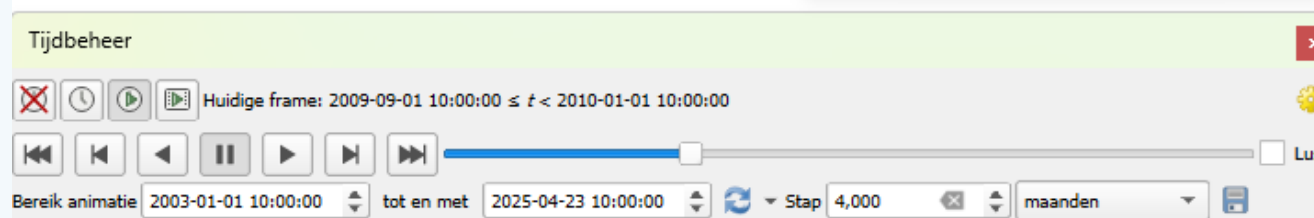
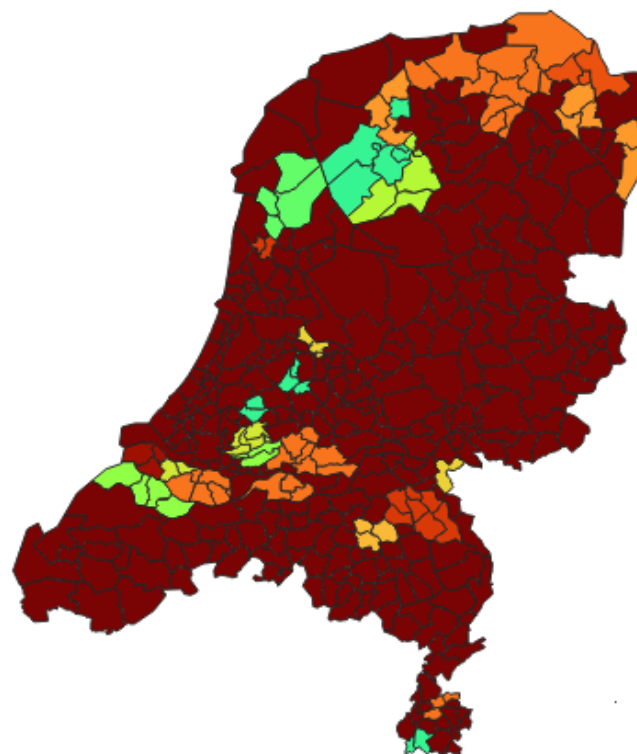
Place or Geometry

- Use place element when special FG-data types are used or when CRS is not WGS84
- Use geometry element when CRS is WGS84 and no special FG-data types are used
- When place and geometry are both used, they should not be identical

Temporal data

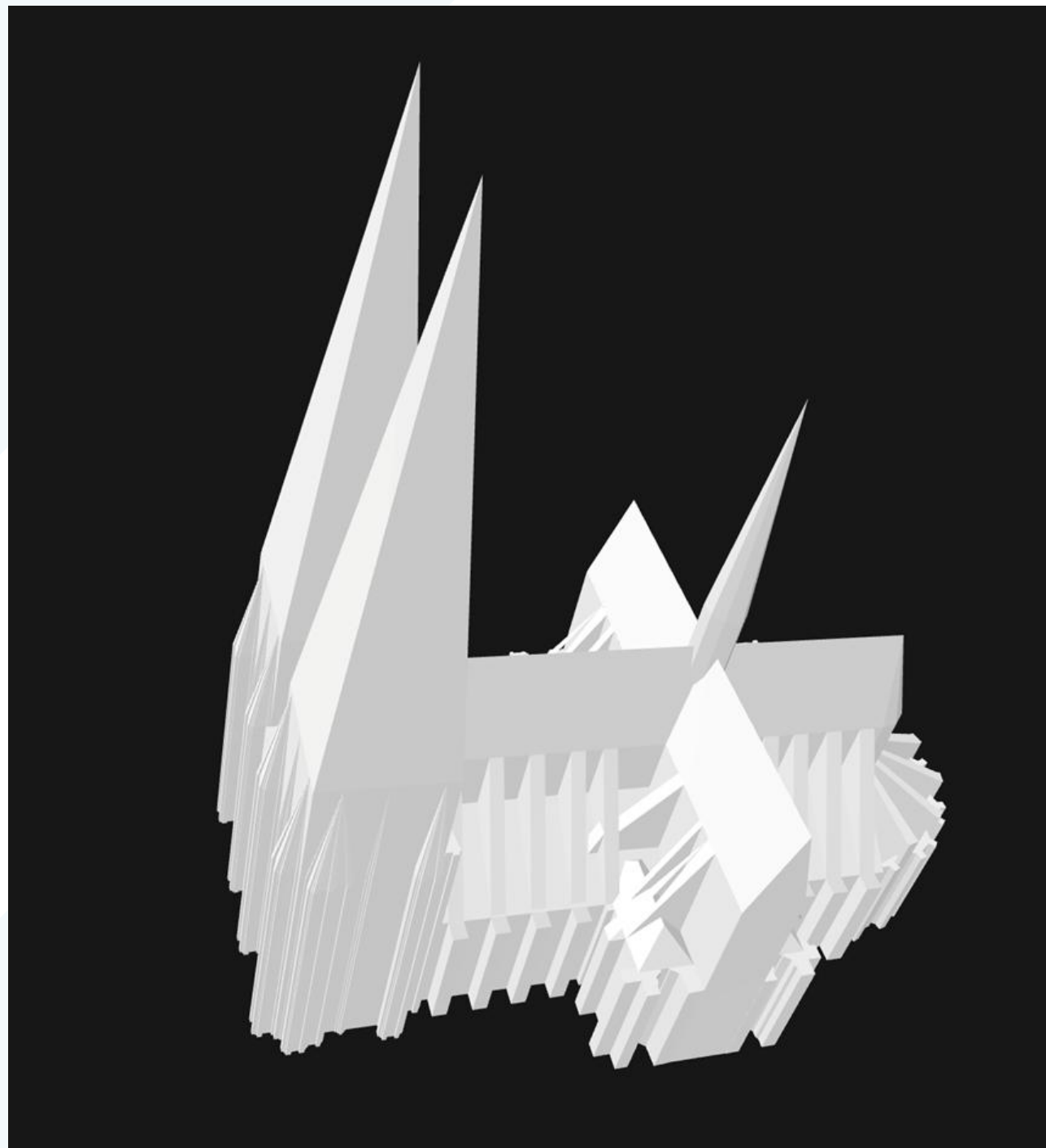
- Use **time**-element
- Instants: "time": {"instant": "2022-10-21T18:00:00Z"}
 - Date: "2002-12-31"
 - Timestamp: "2002-12-31T23:00:00Z"
- Intervals: "time": { "interval": ["2002-12-31T23:00:00Z", "2003-12-31T23:00:00Z"] } }

Temporal data



3D

- Polyhydron
- Multi-Polyhydron



3D

- Prism (polygon or circle with upper and lower limit) 2,5 D
- Muti-prism



Complex geometry

- **CircularString**
 - a sequence of an odd number of points with a minimum of three points
- **CompoundCurve**
 - a sequence of circular strings and/or line strings
- **CurvePolygon**
 - a sequence of closed rings, build up by line strings, circular strings or line strings
 - The first ring is the exterior boundary and, all other rings are interior boundaries
- **MultiCurve**
 - a set of line strings, circular strings or compound curves
- **MultiSurface**
 - a set of polygons or curve polygons

Complex geometry: CircularString

```

1 {
2   "conformsTo": [
3     "http://www.opengis.net/spec/json-fg-1/1.0/conf/core" ,
4     "http://www.opengis.net/spec/json-fg-1/1.0/conf/circular-arcs"
5   ],
6   "coordRefSys": "http://www.opengis.net/def/crs/OGC/0/CRS84",
7   "type": "CircularString",
8   "coordinates": [
9     [4.5368482, 51.5913298],
10    [4.5371346, 51.591267],
11    [4.5373782, 51.591154]
12  ]
13 }
```

Arc

Circle

```

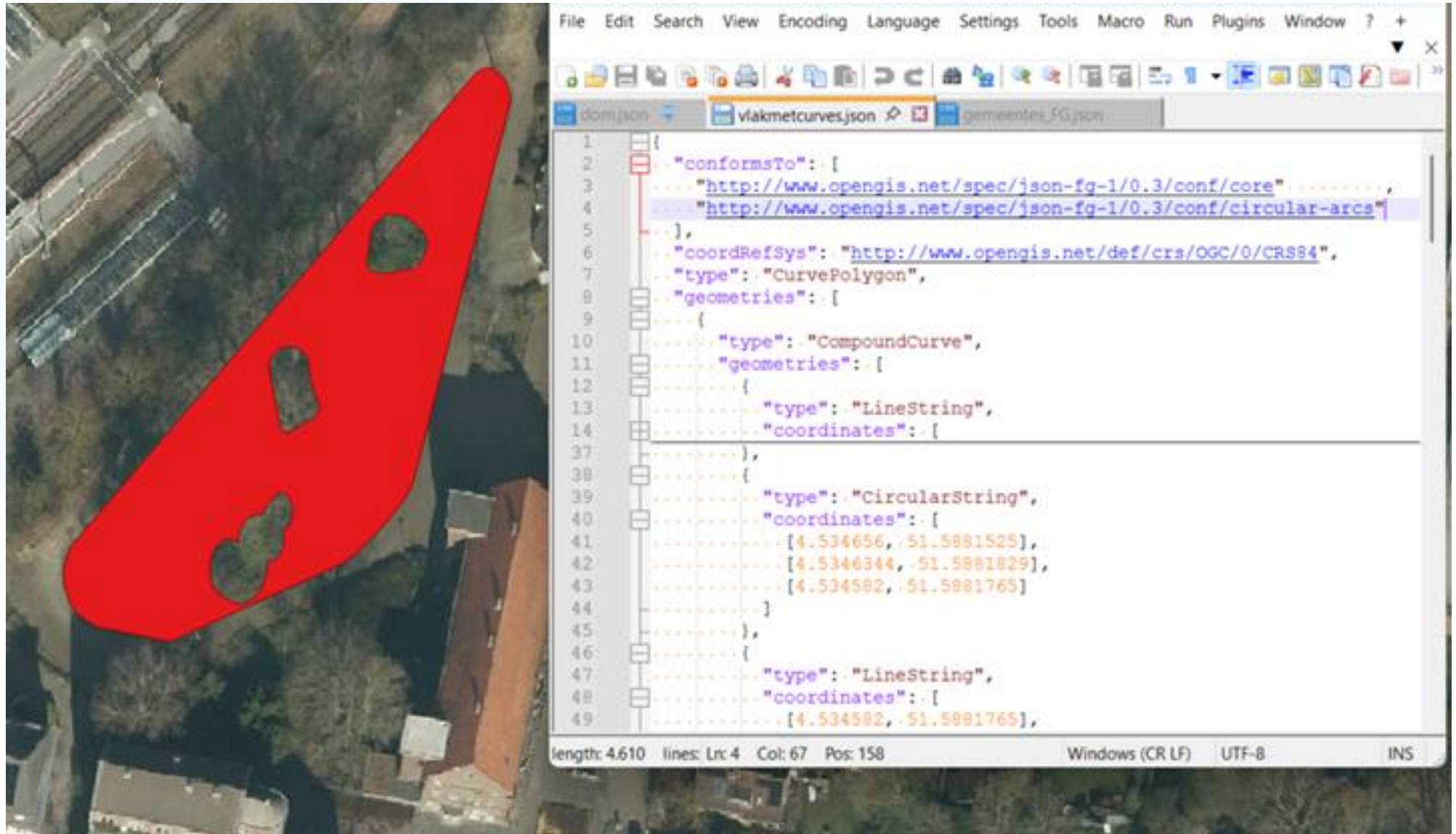
1 {
2   "conformsTo": [
3     "http://www.opengis.net/spec/json-fg-1/1.0/conf/core" ,
4     "http://www.opengis.net/spec/json-fg-1/1.0/conf/circular-arcs"
5   ],
6   "coordRefSys": "http://www.opengis.net/def/crs/OGC/0/CRS84",
7   "type": "CircularString",
8   "coordinates": [
9     [4.5390810, 51.5902305],
10    [4.5390200, 51.5902275],
11    [4.5390248, 51.5901896],
12    [4.5390857, 51.5901925],
13    [4.5390810, 51.5902305]
14  ]
15 }
```

Complex geometry:

CompoundCurve

```
1 {
2   "conformsTo": [
3     "http://www.opengis.net/spec/json-fg-1/1.0/conf/core"
4     "http://www.opengis.net/spec/json-fg-1/1.0/conf/circular-arcs"
5   ],
6   "coordRefSys": "http://www.opengis.net/def/crs/OGC/0/CRS84",
7   "type": "CompoundCurve",
8   "geometries": [
9     {
10      "type": "LineString",
11      "coordinates": [
12        [4.5362932, 51.5906989],
13        [4.5362737, 51.5906938]
14      ]
15    },
16    {
17      "type": "CircularString",
18      "coordinates": [
19        [4.5362737, 51.5906938],
20        [4.5362567, 51.5906829],
21        [4.5362591, 51.5906678]
22      ]
23    },
24    {
25      "type": "LineString",
26      "coordinates": [
27        [4.5362591, 51.5906678],
28        [4.5363324, 51.5906866]
29      ]
30    },
31    {
32      "type": "CircularString",
33      "coordinates": [
34        [4.5363324, 51.5906866],
35        [4.5363173, 51.5906983],
36        [4.5363037, 51.5906992]
37      ]
38    }
39  ]
40 }
```


Complex geometry: CurvePolygon



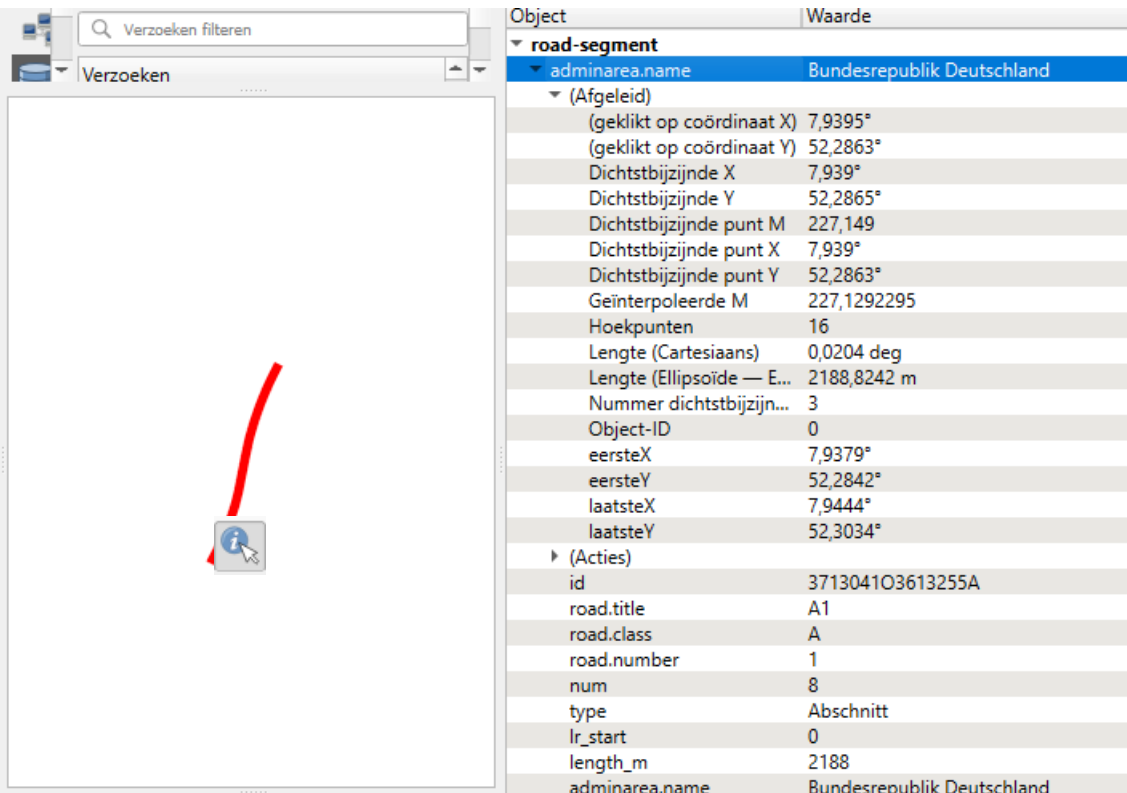
Measure

- Linear referencing:



Measure

```
[{"type": "Feature",
  "id": "371304103613255A",
  "conformsTo": [
    "http://www.opengis.net/spec/json-fg-1/1.0/conf/core",
    "http://www.opengis.net/spec/json-fg-1/1.0/conf/measures"
  ],
  "coordRefSys": "http://www.opengis.net/def/crs/OGC/0/CRS84",
  "measures": {
    "enabled": true,
    "unit": "km",
    "description": "Kilometers along the road based on the kilometer markers. Positions between two physical markers are interpolated."
  },
  "geometry": null,
  "place": {
    "type": "LineString",
    "coordinates": [
      [7.9379077, 52.2841795, 227.396],
      [7.9386618, 52.2856218, 227.227],
      [7.9389695, 52.2862999, 227.149],
      [7.9395120, 52.2876169, 226.998],
      [7.9397562, 52.2883001, 226.920],
      [7.9401802, 52.2896330, 226.769],
      [7.9405226, 52.2910055, 226.614],
      [7.9412150, 52.2945101, 226.222],
      [7.9415283, 52.2959768, 226.057],
      [7.9416385, 52.2964391, 226.005],
      [7.9421299, 52.2981305, 225.814],
      [7.9426214, 52.2995303, 225.655],
      [7.9429592, 52.3003909, 225.556],
      [7.9434566, 52.3015636, 225.422],
      [7.9438507, 52.3024014, 225.325],
      [7.9443588, 52.3034069, 225.208]
    ]
  },
  "time": { "instant": "2022-10-21T18:00:00Z" },
  "properties": {
    "road.title": "A1",
    "road.class": "A",
    "road.number": 1,
    "num": 8,
    "type": "Abschnitt",
    "lr_start": 0,
    "length_m": 2188,
    "adminarea.name": "Bundesrepublik Deutschland",
    "adminarea.knz": "000000000000",
    "netzstand": "2022-10-21T18:00:00Z"
  }
}]
```



The screenshot shows a GIS application interface. On the left, a map displays a red line segment representing a road. A red arrow points to a blue information icon on the map. On the right, a table lists attributes for the selected object, 'road-segment'.

Object	Waarde
road-segment	
adminarea.name	Bundesrepublik Deutschland
(Afgeleid)	
(geklikt op coördinaat X)	7,9395°
(geklikt op coördinaat Y)	52,2863°
Dichtstbijzijnde X	7,939°
Dichtstbijzijnde Y	52,2865°
Dichtstbijzijnde punt M	227,149
Dichtstbijzijnde punt X	7,939°
Dichtstbijzijnde punt Y	52,2863°
Geïnterpoleerde M	227,1292295
Hoekpunten	16
Lengte (Cartesiaans)	0,0204 deg
Lengte (Ellipsoïde — E...	2188,8242 m
Nummer dichtstbijzijn...	3
Object-ID	0
eersteX	7,9379°
eersteY	52,2842°
laatsteX	7,9444°
laatsteY	52,3034°
(Acties)	
id	371304103613255A
road.title	A1
road.class	A
road.number	1
num	8
type	Abschnitt
lr_start	0
length_m	2188
adminarea.name	Bundesrepublik Deutschland
adminarea.knz	000000000000
netzstand	21-10-2022 18:00:00 (UTC)

Schema

- featureType for whole json-file or per feature

```
{
  "type": "FeatureCollection",
  "conformsTo": [
    "http://www.opengis.net/spec/json-fg-1/0.3/conf/core",
    "http://www.opengis.net/spec/json-fg-1/0.3/conf/types-schemas"
  ],
  "featureType": "gemeentes",
  "coordRefSys": "http://www.opengis.net/def/crs/EPSSG/0/28992",
  "features": [
    {
      "type": "Feature",
      "properties": {
        "code": "0258",
        "naam": "Kesteren",
        "begingeldigheid": "2002-12-31",
        "eindgeldheid": "2003-03-31"
      }
    },
    {
      "type": "Feature",
      "properties": {
        "code": "0178",
        "naam": "Rijssen",
        "begingeldigheid": "2002-12-31",
        "eindgeldheid": "2003-12-31"
      }
    }
  ]
}
```

```
1 {
2   "type": "FeatureCollection",
3   "coordRefSys": "http://www.opengis.net/def/crs/EPSSG/0/5555",
4   "conformsTo": [
5     "http://www.opengis.net/spec/json-fg-1/0.3/conf/core",
6     "http://www.opengis.net/spec/json-fg-1/0.3/conf/types-schemas"
7   ],
8   "features": [
9     {
10      "type": "Feature",
11      "featureType": "Building",
12      "id": "1672388",
13      "geometry": null,
14      "properties": {
15      }
16    },
17    {
18      "type": "Feature",
19      "featureType": "BuildingPart",
20      "id": "2300938",
21      "geometry": null,
22      "properties": {
23        "place": {
24          "type": "Polyhedron",
25          "coordinates": [
26            [
27              [
28                [
29                  [
30                    [
31                      [
32                        [
33                          [
34                            [
35                      ]
36                    ]
37                  ]
38                ]
39              ]
40            ]
41          ]
42        ]
43      ]
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97  ]
98  ]
99  ]
100 ]
}
```

Profiles

- HTTP: in HTTP header "Link" using the link relation "profile". (value of link = URI of profile)
- Or in JSON-FG document
- *RFC 7946 (GEOJSON)*
Link: <<http://www.opengis.net/def/profile/OGC/0/rfc7946>>; rel="profile"
- *JSON-FG*
Link: <<http://www.opengis.net/def/profile/OGC/0/jsonfg>>; rel="profile"
- *JSON-FG with improved support for GeoJSON readers*
Link: <<http://www.opengis.net/def/profile/OGC/0/jsonfg-plus>>; rel="profile"

Validations

Geonovum OGC Checker

geonovum.github.io/ogc-checker/#/json-fg

JIRA Exact Online NGR INSPIRE Geoportal Geonovum nieuws Geonovum IMGeluid IMEV inspire-handreiking

Geonovum OGC Checker: JSON-FG

Enter URL to load a document from remote location... Load

JSON-FG

1 {
2 "type": "FeatureCollection",
3 "coordRefSys": "http://www.opengis.net/def/crs/EPSSG/0/5555",
4 "conformsTo": [
5 "http://www.opengis.net/spec/json-fg-1/0.3/conf/core",
6 "http://www.opengis.net/spec/json-fg-1/0.3/conf/types-schemas",
7 "http://www.opengis.net/spec/json-fg-1/0.3/conf/polyhedra"
8],
9 "features": [
10 {
11 "type": "Feature",
12 "featureType": "Building",
13 "id": 1672388,
14 "geometry": null,
15 "properties": {
16 "gml_id": "DENW37AL1000qmxq",
17 "creationDate": "2019-12-27",
18 "externalReferences": [
19 {
20 "name": "DENW37AL1000qmxq",
21 "informationSystem": "https://repository.gdi-de.org/schemas/adv/ci"
22 }
23],
24 "function": "31001_3041",
25 "address": [
26 {
27 "ThoroughfareName": "Domkloster",
28 "ThoroughfareNumber": "4",
29 "LocalityName": "Köln",
30 "CountryName": "Germany"
31 }
32],
33 "gemeindeschlüssel": "05315000",
34 "consistsOfBuildingParts": [
35 {
36 "\$ref": "#/features/1"
37 },
38 {
39 "\$ref": "#/features/2"
40 }
41]
42 }
43],
44 }
45 }

JSON YAML

[json-fg-1/1.0/conf/core] Found 1 issue(s): 1 error(s), 0 warning(s), 0 hint(s), 0 info.

Errors (1)

[/req/core/schema-valid] The JSON object SHALL validate against the JSON Schema of a JSON-FG root object. "conformsTo" property must contain at least 1 valid item(s).
Show in editor Documentation

[json-fg-1/1.0/conf/polyhedra] Found 1 issue(s): 1 error(s), 0 warning(s), 0 hint(s), 0 info.

Errors (1)

[/req/polyhedra/metadata] The "conformsTo" member of a JSON-FG root object that contains any of the polyhedron geometry types SHALL include the value
① "http://www.opengis.net/spec/json-fg-1/1.0/conf/polyhedra".
Show in editor Documentation

[json-fg-1/1.0/conf/prisms] Found 1 issue(s): 1 error(s), 0 warning(s), 0 hint(s), 0 info.

Errors (1)

[/req/prisms/metadata] The "conformsTo" member of a JSON-FG root object that contains any of the prism geometry types SHALL include the value
① "http://www.opengis.net/spec/json-fg-1/1.0/conf/prisms".
Show in editor Documentation

Get started with JSON-FG yourself

- Exercises are available on Github
 - <https://github.com/Geonovum/ogc-api-workshops/>
- View a JSON-FG file in browser or JSON-editor
 - CRS
 - 3D (<https://dev.3dgi.xyz/jsonfg-viewer/#>)
 - Curves
 - Measures
- Validate JSON-FG
- Lunch break
- Brainstorm pilot project with data exchange
- Load JSON-FG into QGIS
- Create a time-slider on a JSON-FG in QGIS
- Create a 3D-scene in QGIS
- Inspect a road segment with measures in QGIS

Lessons learned during the workshop

1. An overview of the JSON-FG standard
2. Dealing with time and other CRS's
3. The extra data types:
 1. 3D
 2. Curves
 3. Measures
4. Validation
5. Loading JSON-FG into GIS software.

Follow up

- Presentations and handson remain available on Github.
- Ask your questions via the [GeoForum](#).
- Feedback from participants?
- More workshops to follow, see [Geonovum agenda](#).
- More information:
 - <https://docs.ogc.org/is/21-045r1/21-045r1.html>
 - <https://github.com/opengeospatial/ogc-feat-geo-json>

Thank you!

Geonovum

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